

NON-LOCAL SURFACE KNOTS AND SLICE DISKS

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ABSTRACT. We show the existence of non-local knotted surfaces in any 4-manifold X with $b_1(X) > 0$. To do so, we investigate how Alexander invariants can be modified to obstruct the locality of surface knots and slice disks.

1. OVERVIEW

This paper is primarily concerned with compact, connected, oriented surfaces J embedded in a smooth 4-manifold X , called *knotted surfaces* or *surface knots*. When the underlying surface of J is S^2 , we also call J a 2-knot. We will work throughout with the assumption that all manifolds and embeddings are smooth, and equivalence is considered up to smooth isotopy. In particular, we are interested in the property of *locality* of these objects:

Definition 1.1. In the classical setting, a knot K in a 3-manifold M is *local* if there is a 3-ball B^3 embedded in M such that $K \subset B^3$. Similarly, a surface knot J in a 4-manifold X is *local* if it is contained in a 4-ball B^4 embedded in X .

The main goal of this paper is to prove the following:

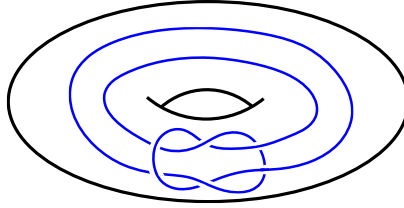
Theorem 1.1. *For any compact smooth 4-manifold X with $b_1(X) > 0$ and any compact connected surface Σ , there exists a null-homotopic surface knot J in X which is not isotopic to any local surface knot $J' \subset B^4 \subset X$, where J, J' are both embeddings of the surface Σ .*

The argument we present leads to a potentially interesting corollary regarding the classical notion of sliceness. Suppose X is a 4-manifold with nonempty boundary $M = \partial X$. Recall that a knot K in M is *slice* in X if it is the boundary of a disk embedded in X , which we'll call a *slice disk*.

Corollary 1.1. *Let $K \subset B^3 \subset M$ be a local knot which is slice in B^4 , and suppose M has $b_1(M) > 0$. Then K admits a slice disk D which is not isotopic to any local slice disk $D' \subset B^4 \subset M \times I$.*

This discussion of non-local surface knots can be thought of as a one-higher-dimensional analogue of some prominent questions about classical non-local knots and concordance. The classical result corresponding to our question might be phrased as follows: in a 3-manifold M where $\pi_1(M) \neq 1$, there exists a null-homotopic knot K such that K is not isotopic to any local knot $K' \subset B^3 \subset M$. A well-known method (see, for instance [Cel18]) of constructing classical null-homotopic non-local knots is as follows: let J be a non-local knot in M which is not null-homologous. Consider the ribbon pattern $P \subset S^1 \times D^2$ as in Figure 1. Let $K = J_P$ be the satellite of J with pattern P in M . Indeed, K is null-homotopic and is an example of a non-local knot in M — if there was a 3-ball B such that $K \subset B \subset M$, one can isotope B to contain J , contradicting the non-locality of J .

The analogous argument breaks down when we generalize to the case of 2-knots. Although we may be able to run a similar satellite construction in a 4-manifold X with $\pi_2(X) \neq 1$,

FIGURE 1. The ribbon pattern $P \subset S^1 \times D^2$

it is not clear how to manually isotope a 4-ball B in the same way as above. It is even less obvious how this would work for surface knots of higher genus. For these reasons, we need to develop other means of investigating the locality of surface knots. Our general strategy will be to use *Alexander invariants*, like the multi-variable *Alexander polynomial* defined for surface links, to obstruct locality. We will associate a variant $\Delta_J^X(s, t)$ of this multi-variable polynomial to a surface knot $J \subset X$ and a distinguished generator of $H_1(X \setminus \nu(J))$ such that, if J is local, then $\Delta_J^X(s, t) = \Delta_J(t)$, where $\Delta_J(t)$ is the usual Alexander polynomial for J in S^4 .

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2. PRELIMINARIES

2.1. Exteriors of 2-Links. We first recall some basic ways to think about surface knots and their exteriors; for a detailed exposition, see [GS99]. In general, a surface knot J can be visualized using *band diagrams*. We consider the cross sections of J living in concentric S^3 's in S^4 , which we'll parametrize as $J \subset S^3 \times [0, 2]$.

Theorem 2.1 ([Lom81], [KSS82]). *For any surface knot J in S^4 , there exists a surface knot J' such that*

- (1) J is isotopic to J' .
- (2) If we parametrize $J' \subset S^3 \times [0, 2]$ as before, the height function $f : J' \rightarrow [0, 2]$ given by $J' \cap S^3 \times \{t\} \mapsto t$ for all $t \in [0, 2]$ is Morse.
- (3) f has only finitely many critical points.
- (4) All index i critical points of f lie in $S^3 \times \{i\}$.

Definition 2.1. A *banded unlink diagram* or *band diagram* is a pair (L, v) where L is an unlink in S^3 and v is a set of bands (2-dimensional 1-handles) attached to L such that the surgery L_v of L along v is an unlink.

Given a knotted surface J in S^4 , we can obtain a banded unlink such that, up to isotopy,

- (1) $J \subset S^3 \times [0, 2] \subset S^4$.
- (2) $J \cap S^3 \times \{0\}$ is the disjoint union of the obvious disks bounded by L .
- (3) $J \cap S^3 \times \{t\}$ where $0 < t < 1$ is the link L .
- (4) $J \cap S^3 \times \{1\}$ is the union $L \cup v$.
- (5) $J \cap S^3 \times \{t\}$ where $1 < t < 2$ is the surgery L_v .
- (6) $J \cap S^3 \times \{2\}$ is the disks bounded by L_v .

Theorem 2.2 ([KSS82] and [KSS83]). *If (L, v) is a band diagram of J and J' , then J, J' are smoothly isotopic in S^4 .*

Example. Relevant to our computations will be the band diagrams for *spun knots* and *twist-spun knots*, introduced by Artin in [Art25] and Zeeman in [Zee65], respectively. Our running example will be the spun trefoil, which has a band diagram

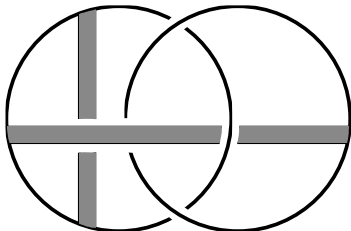


FIGURE 2. Band diagram for the spun trefoil

From a band diagram (L, v) of a knotted surface J , it is straightforward to draw a handle diagram for its exterior $X_J = S^4 \setminus \nu(J)$. As in [GS99], taking the complement of a 2-dimensional k -handle of J is equivalent to adding a $(k + 1)$ -handle to X_J . Thus, in our diagram, we can replace the unknotted components of L with dotted circles representing 1-handles and the bands in v with 0-framed 2-handles running around the 1-handles once on either end.

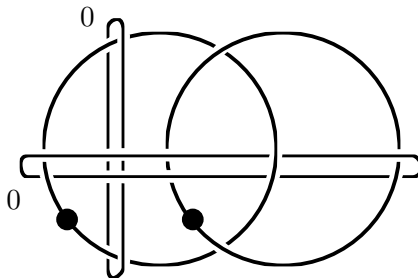


FIGURE 3. Handle diagram for the exterior of the spun trefoil

2.2. Alexander Invariants of 2-Links. As is the case for classical knots and links, the algebraic topology of infinite abelian covers for the exteriors of surface knots yields several useful invariants, called *Alexander invariants*. To obstruct locality, we will make use of the *Alexander module*, *Alexander ideals*, and the *Alexander polynomial* associated to surface knots. We will define these in this section; for general references, see Chapter 7 of [Kaw96] for the case of classical links and [Jos20] for Alexander invariants of surface knots. Although [Kaw96] is one dimension down, most of the arguments and results apply without much modification.

Let us consider an r -component link of surface knots $L = L_1 \cup \cdots \cup L_r$ in S^4 , so that each L_i is a smoothly embedded surface knot disjoint from any other L_j . If the underlying surface of each L_i were a 2-knot, this would be a *2-link*. Let X_L be the exterior of L . For convenience, let $\pi = \pi_1(X_L)$ and t_i be the homology class in $H_1(X_L) = H_1(S^4 \setminus \nu(L))$ represented by the meridian of L_i where $1 \leq i \leq r$. Then $H_1(X_L)$ is a free abelian group

of rank r generated by $t_1, \dots, t_r$. Let $\mathbf{a} : \pi = \pi_1(X_L) \twoheadrightarrow H_1(X_L)$ be the Hurewicz map. The covering space X_L^∞ over X_L corresponding to the subgroup $\ker(\mathbf{a}) = [\pi, \pi]$ of π is the *universal abelian covering space* of X . Since $H_1(X_L)$ acts on X_L^∞ as the deck transformation group, we can regard $H_1(X_L)$ as a module over the integral group ring $\mathbb{Z}H_1(X_L)$, which we can identify with the ring of multi-variable Laurent polynomials $\Lambda_r = \mathbb{Z}[t_1^{\pm 1}, \dots, t_r^{\pm 1}]$. So, $H_1(X_L^\infty)$ has the additional structure of a Λ_r -module. Moreover, if $p : X_L^\infty \rightarrow X_L$ is our covering map and b some base point in X , then $H_1(X_L^\infty, p^{-1}(b))$ also has the structure of a Λ_r -module.

Definition 2.2. The *link module* of L is the Λ_r -module $H_1(X_L^\infty)$, and the *Alexander module* of L is the Λ_r -module $H_1(X_L^\infty, p^{-1}(b))$, which we'll denote $\mathcal{A}(L)$.

From this, there are a number of ways to define polynomial invariants analogous to that of the classical multi-variable Alexander polynomial. One option is as follows: suppose we have an $m \times n$ presentation matrix A for $\mathcal{A}(L)$. The $(n-i) \times (n-i)$ minor determinants of A generate an ideal of Λ_r , called the *i -th elementary ideal* of $\mathcal{A}(L)$ and which we'll denote by $\mathcal{E}_i\mathcal{A}(L)$. The greatest common divisor of these generators for $\mathcal{E}_i\mathcal{A}(L)$ is a multi-variable Laurent polynomial which we'll denote by $\Delta_i\mathcal{A}(L)$.

Definition 2.3. The *Alexander polynomial* $\Delta_L(t_1, \dots, t_r)$ of a link of surface knots L is the multi-variable Laurent polynomial $\Delta_2\mathcal{A}(L)$.

A convenient way of computing a presentation matrix for $\mathcal{A}(L)$ (and therefore the Alexander polynomial) is *Fox's free differential calculus*, which we'll quickly review. For more on the case for classical knots, see Chapter 7 of [Kaw96]. Suppose $\langle x_1, \dots, x_n \mid r_1, \dots, r_m \rangle$ is a presentation for the link group π , and let F_n be the free group with basis $x_1, \dots, x_n$. For all $1 \leq k \leq n$, there is a unique map $\partial/\partial x_k : F_n \rightarrow \mathbb{Z}F_n$ determined by the conditions that

$$\frac{\partial x_i}{\partial x_k} = \delta_{ik} \quad \text{and} \quad \frac{\partial(uv)}{\partial x_k} = \frac{\partial u}{\partial x_k} + u \frac{\partial v}{\partial x_k}.$$

Uniqueness of this map can be shown inductively; note the base cases $\partial 1/\partial x_k = 0$ and $\partial x_k^{-1}/\partial x_k = -x_k^{-1}$. If $\mathbf{q} : F_n \rightarrow \pi$ is the universal quotient map, then we get the induced maps on group rings

$$\mathbb{Z}F_n \xrightarrow{\partial/\partial x_k} \mathbb{Z}F_n \xrightarrow{\mathbf{q}} \mathbb{Z}\pi \xrightarrow{\mathbf{a}} H_1(X_L) = \mathbb{Z}\langle t_1, \dots, t_r \rangle.$$

This gives rise to a chain complex

$$\Lambda_r^m \xrightarrow{\partial_2} \Lambda_r^n \xrightarrow{\partial_1} \Lambda_r \rightarrow 0,$$

such that ∂_1 is the represented by the matrix $(\mathbf{a}\mathbf{q}(x_k) - 1)$ where $1 \leq k \leq n$ and ∂_2 by the matrix

$$A = \left(\mathbf{a}\mathbf{q} \left(\frac{\partial r_i}{\partial x_j} \right) \right)$$

where $1 \leq i \leq m$ and $1 \leq k \leq n$. As in [Kaw96], we can identify $\mathcal{A}(L)$ and $H_1(X_L^\infty)$ with $\text{coker}(\partial_2)$ and $\ker(\partial_1)/\text{im}(\partial_2)$, respectively, and so we find that A is a presentation matrix for the Alexander module.

Theorem 2.3 ([Kaw96]). *Although, in general, a presentation matrix for $\mathcal{A}(L)$ is not unique, it is unique up to a finite series of the following moves:*

- (1) *Performing standard row and column operations over Λ_r .*
- (2) *Adding a row $A \leftrightarrow \begin{pmatrix} A \\ * \end{pmatrix}$, where $*$ is a Λ_r -linear combination of the rows of A .*

(3) *Stabilization*, where $A \leftrightarrow \begin{pmatrix} A & 0 \\ 0 & 1 \end{pmatrix}$ where the bottom left entry is a row of zeroes.

This gives rise to a number of other invariants coming from $\mathcal{A}(L)$, for example the *rank* of $\mathcal{A}(L)$, defined to be the rank of the $\mathbb{Q}\Lambda_r$ -vector space $\mathcal{A}(L) \otimes_{\Lambda_r} \mathbb{Q}\Lambda_r$. Using Theorem 2.3, it is relatively straightforward to see that the elementary ideals, Alexander polynomial, and these other invariants do not depend on the choice of presentation matrix.

We make a quick technical note on the difference between using the Alexander module $\mathcal{A}(L)$ and the link module $H_1(X_L^\infty)$ to define our polynomial invariants. In the case where our surface knot K is connected, there is nothing to distinguish, since $\mathcal{A}(K) \cong H_1(X_L^\infty)$. In the case of a surface link with more than one component, this is no longer true. The primary source [Jos20] we use for basic theorems on the relationship between operations on surface knots and the Alexander polynomial tends to use the link module, and so these theorems do not necessarily apply in our context of r -component surface links. However, we have the following theorem:

Theorem 2.4 ([Kaw96]). *For a surface link L with r -components, we have that*

$$\Delta_2 \mathcal{A}(L) = \Delta_L(t_1, \dots, t_r) = \Delta_1(H_1(X_L^\infty)).$$

In fact, this holds for $\Delta_{i+1} \mathcal{A}(L)$ and $\Delta_i H_1(X_L^\infty)$ for all $i \geq 1$.

This is helpful because it is easy to compute a presentation matrix for $\mathcal{A}(L)$, hence to compute $\Delta_2 \mathcal{A}(L)$, and so we can readily apply the results from [Jos20] regarding $\Delta_1(H_1(X_L^\infty))$.

2.3. Operations on Surface Knots. Before shifting focus to the proof of Theorem 1.1, we note some relevant results about the Alexander polynomial for special cases of surface knots. In particular, given surface knots J and K in a 4-manifold X , we're interested in the Alexander polynomial of the band sum $J \natural_\alpha K$ along an arc α (when this operation is well defined) and of a split link consisting of J and K . We first recall the definition we will use for band sums of surface knots.

Definition 2.4. Let J and K be disjoint surface knots in a 4-manifold X and α an arc in X with endpoints $p \in J$ and $q \in K$. Consider the tubular neighborhood $B = \nu(\alpha)$ which intersects J and K at the disks $b_1 = J \cap B$ and $b_2 = K \cap B$. Then the *band sum* $J \natural_\alpha K$ is the knot

$$J \cup K \cup (\partial B \setminus (\text{int } b_1 \cup \text{int } b_2)).$$

Note that even in S^4 , the surface knot $J \natural_\alpha K$ depends on the choice of α . However, it is always possible to choose α so that $J \natural_\alpha K$ satisfies the algebraic properties necessary for our argument of Theorem 1.1.

Proposition 2.1. *Suppose that*

$$\pi_1(X \setminus \nu(J)) \quad \text{and} \quad \pi_1(X \setminus \nu(K))$$

are presented by

$$\langle x_1, \dots, x_n \mid r_1, \dots, r_k \rangle \quad \text{and} \quad \langle y_1, \dots, y_m \mid s_1, \dots, s_\ell \rangle,$$

respectively. Then there exists an arc α in X such that

$$\pi_1(X \setminus \nu(J \natural_\alpha K)) = \langle x_1, \dots, x_n, y_1, \dots, y_m \mid x_1 y_1^{-1}, r_1, \dots, r_k, s_1, \dots, s_\ell \rangle.$$

This can be seen from a standard relative handle manipulation argument. Since the Alexander invariants of a surface knot depend only on the fundamental group of its exterior, Proposition 2.1 is all we need to reason about the invariants of the band sum.

Proposition 2.2. *If J and K are surface knots in S^4 and α is an arc as in Proposition 2.1, then*

$$\Delta_{J \natural_\alpha K}(t) = \Delta_J(t) \Delta_K(t).$$

Proof. A proof using the above free calculus method can be found in the proof of Proposition 2.10 in [Jos20]. $\square$

We will revisit Alexander invariants of band sums when we define them in a general 4-manifold X .

Proposition 2.3. *If $L = J \cup K$ is a two-component split link of surface knots, then*

$$\Delta_L(s, t) = \Delta_J(s) \Delta_K(t).$$

Proof. By Seifert-Van Kampen, we find that

$$\pi_1(S^4 \setminus \nu(L)) = \pi_1(S^4 \setminus \nu(J)) \oplus \pi_1(S^4 \setminus \nu(K)).$$

This can be seen by choosing a splitting ball $B^4 \subset S^4$ such that $J \subset B$, so that $S^4 \setminus B$ is another copy of B^4 such that $K \subset S^4 \setminus B$. So, if

$$\pi_1(S^4 \setminus \nu(J)) \quad \text{and} \quad \pi_1(S^4 \setminus \nu(K))$$

are presented by

$$\langle x_1, \dots, x_n \mid r_1, \dots, r_k \rangle \quad \text{and} \quad \langle y_1, \dots, y_m \mid s_1, \dots, s_\ell \rangle,$$

respectively, then it follows that

$$\pi_1(S^4 \setminus \nu(L)) = \langle x_1, \dots, x_n, y_1, \dots, y_m \mid r_1, \dots, r_k, s_1, \dots, s_\ell \rangle.$$

Thus a presentation matrix for the Alexander module $\mathcal{A}(L)$ is the block matrix

$$\begin{pmatrix} \frac{\partial r_1}{\partial x_1} & \cdots & \frac{\partial r_1}{\partial x_n} & 0 & \cdots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ \frac{\partial r_k}{\partial x_1} & \cdots & \frac{\partial r_k}{\partial x_n} & 0 & \cdots & 0 \\ 0 & \cdots & 0 & \frac{\partial s_1}{\partial y_1} & \cdots & \frac{\partial s_1}{\partial y_m} \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & \frac{\partial s_\ell}{\partial y_1} & \cdots & \frac{\partial s_\ell}{\partial y_m} \end{pmatrix}.$$

We find then that the greatest common divisor $\Delta_L(s, t)$ of the generators of $\mathcal{E}_2 \mathcal{A}(L)$ is the product $\Delta_J(s) \Delta_K(t)$. $\square$

Lemma 2.1. *If U is an unknotted surface knot in S^4 — that is it bounds an embedded handlebody — then $\Delta_U(t) = 1$.*

Proof. Since U is unknotted,

$$\pi_1(S^4 \setminus \nu(U)) \cong \mathbb{Z} = \langle \mu \rangle$$

where μ is the meridian of U . Having no relations, $\mathcal{A}(U)$ is presented by the matrix (1) , and so $\Delta_U(t) = 1$. $\square$

3. EXISTENCE OF NON-LOCAL SURFACE KNOTS

Using these invariants, we can now move towards the proof of the existence of non-local surface knots and slice disks. As in the statement of Theorem 1.1, let X be a 4-manifold with $b_1(X) > 0$ and let Σ be an abstract compact connected surface. Our general argument will proceed as follows: first, we construct a non-local 2-knot in X . To do so, we construct a non-local 2-knot J in $S^1 \times B^3$, which we can then embed into X in an interesting way. We need to argue then that J is actually non-local in both spaces. This is where we can adapt our discussion of Alexander invariants to this more general setting: we will define a two-variable Alexander polynomial $\Delta_J^{S^1 \times B^3}(s, t)$ associated to J in $S^1 \times B^3$, and if this polynomial does not agree with $\Delta_J(t)$, then we can conclude J is non-local in $S^1 \times B^3$. Similarly, we will extend $\Delta^{S^1 \times B^3}(s, t)$ to an Alexander polynomial associated to the embedding of J in X , and we will use this to show non-locality.

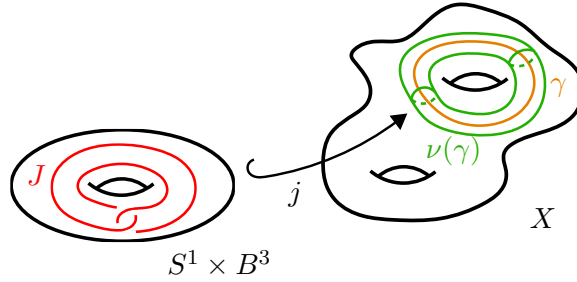


FIGURE 4. A schematic of the non-local 2-knot we construct in $S^1 \times B^3$ and its embedding into X

Once we prove the existence of non-local 2-knots in X , Theorem 1.1 follows relatively quickly. Given a non-local 2-knot J in X and an embedding K of Σ into X , we can form the band sum $J \natural_\alpha K$ along an arc α as in Proposition 2.1. Knowing how the Alexander polynomial acts under such a band sum, we conclude that $J \natural_\alpha K$ is non-local.

Throughout our argument, it will be a common construction to consider some loop γ in a 4-manifold X and identify a neighborhood of γ with the copy of $S^1 \times B^3$ containing our non-local surface knot J . Formally, we consider the surgery $X \setminus \nu(\gamma) \cup S^1 \times B^3$. However, it's important to note that there are exactly two ways to choose this gluing according to a framing on γ , neither of which are canonical. This choice does not affect our strategy due to the following observation:

Proposition 3.1. *If J is a surface knot in $S^1 \times B^3$ and $j_i : \partial(S^1 \times B^3) \rightarrow \partial(X \setminus \nu(\gamma))$ are the gluing diffeomorphisms above for $i = 1, 2$, then $\pi_1(X \setminus \nu(j_1(J)))$ is isomorphic to $\pi_1(X \setminus \nu(j_2(J)))$.*

Proof. Without loss of generality, suppose that j_1 is the identity map on $\partial(S^1 \times B^3) = S^1 \times S^2$ and j_2 is the map which performs a Gluck twist. It can be seen that both of these maps extend over $S^1 \times B^3$ and induce the identity on $\pi_1(S^1 \times S^2)$, hence by Seifert-Van Kampen, the result holds. $\square$

Similar to before, it will turn out that our Alexander invariants only depend on $\pi_1(X \setminus j(J))$ for some embedding j , and so we will choose embeddings $j : S^1 \times B^3 \hookrightarrow X$ without specifying how to make this choice.

3.1. Non-Local 2-Knots in $S^1 \times B^3$. To construct a non-local 2-knot in $S^1 \times B^3$, we can adapt our Alexander invariants to this setting in order to obstruct locality. Let J be a surface knot in $S^1 \times B^3$. Consider an embedding $j : S^1 \times B^3 \hookrightarrow S^4$ given by identifying $S^1 \times B^3$ with a tubular neighborhood of an unknot in S^4 . The complement $S^4 \setminus \nu(j(S^1 \times B^3))$ is diffeomorphic to a copy of $S^2 \times B^2$, and so we can think of $S^1 \times B^3 \setminus \nu(J)$ as $S^4 \setminus \nu(L)$ where $L = U \cup J$ and U is the unknotted 2-sphere sitting at the core of $S^4 \setminus \nu(j(S^1 \times B^3))$.

Definition 3.1. Define the *Alexander polynomial* $\Delta_J^{S^1 \times B^3}(s, t)$ of J in $S^1 \times B^3$ to be

$$\Delta_J^{S^1 \times B^3}(s, t) = \Delta_{L=U \cup J}(s, t),$$

where U is as in the discussion above, and we abuse notation to refer to the embedding $j(J)$ in S^4 as J .

This relates to the locality of J in $S^1 \times B^3$ according to the following lemma:

Lemma 3.1. *If J is a local in $S^1 \times B^3$, then*

$$\Delta_J^{S^1 \times B^3}(s, t) = \Delta_J(t),$$

where $\Delta_J(t)$ is the usual Alexander polynomial of J in S^4 .

Proof. Since J is local in $S^1 \times B^3$, it is also local in S^4 under the embedding j . Thus the link $U \cup J$ is split, and so by Proposition 2.3,

$$\Delta_J^{S^1 \times B^3}(s, t) = \Delta_{U \cup J}(s, t) = \Delta_U(s) \Delta_J(t) = \Delta_J(t),$$

since $\Delta_U(s) = 1$ by Lemma 2.1. □

Also, note how $\Delta_J^{S^1 \times B^3}(s, t)$ acts under band sum.

Proposition 3.2. *If J and K are disjoint surface knots in $S^1 \times B^3$ where K is local and α is an arc as in Proposition 2.1, then*

$$\Delta_{J \natural_\alpha K}^{S^1 \times B^3}(s, t) = \Delta_J^{S^1 \times B^3}(s, t) \Delta_K(t).$$

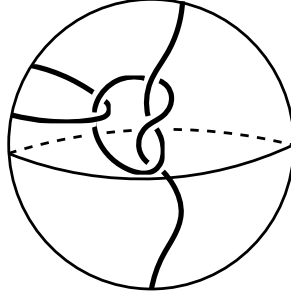
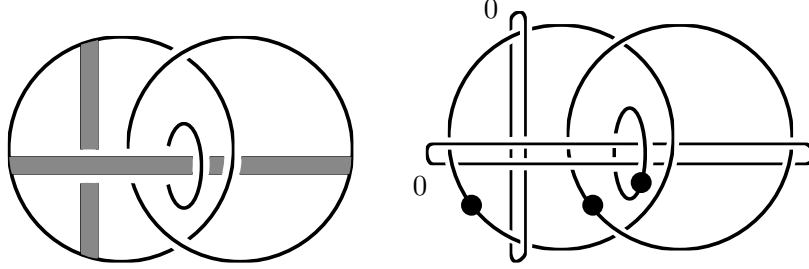
Proof. As before, we can think of $J \natural_\alpha K$ as part of a two-component link in S^4 together with U . Now, following the argument from Proposition 2.10 of [Jos20] and applying Lemma 3.1, we find that

$$\Delta_{J \natural_\alpha K}^{S^1 \times B^3}(s, t) = \Delta_{U \cup J \natural_\alpha K}(s, t) = \Delta_{U \cup J}(s, t) \Delta_K(t) = \Delta_J^{S^1 \times B^3}(s, t) \Delta_K(t),$$

as desired. □

Now, we construct our first non-local 2-knot J in $S^1 \times B^3$. Indeed, J will be a 2-knot very similar to our model example in 2.1. Construct this J as follows: consider the link in B^3 as in Figure 5. Spin this link to obtain a 2-link L in S^4 — we give a band diagram for L and the handle diagram for its exterior in Figure 6. Note that the handle diagram for the exterior of J in S^4 is also a handle diagram for the exterior of J in $S^1 \times B^3$, since $S^4 \setminus \nu(U)$ is diffeomorphic to $S^1 \times B^3$. We can then compute a presentation of the link group of $U \cup J$, so that

$$\pi_1(S^4 \setminus \nu(U \cup J)) = \langle \mu, x, y \mid r = xy^{-1}x^{-1}\mu^{-1}y^{-1}\mu xy \rangle$$

FIGURE 5. The link in B^3 that we spin to obtain $U \cup J$ FIGURE 6. A band diagram for the spun $U \cup J$ and a handle diagram for its exterior

where μ is represented by the meridian of U . Now, using our free differential calculus approach, we can compute $\Delta_{U \cup J}(s, t)$. We find that

$$\begin{aligned} \frac{\partial r}{\partial \mu} &= -xy^{-1}x^{-1}\mu^{-1} + xy^{-1}x^{-1}\mu^{-1}y^{-1} \\ \frac{\partial r}{\partial x} &= 1 - xy^{-1}x^{-1} + xy^{-1}x^{-1}\mu^{-1}y^{-1}\mu + xy^{-1}\mu \\ \frac{\partial r}{\partial y} &= -xy^{-1} - xy^{-1}\mu^{-1}y^{-1} + xy^{-1}x^{-1}\mu^{-1}y^{-1}\mu x \end{aligned}$$

and so $\mathcal{A}(L)$ is presented by the matrix

$$\begin{pmatrix} -t^{-1}s^{-1} + s^{-1} & 1 - t^{-1} + t^{-2} + s & -1 - t^{-1}s^{-1} + t^{-1} \end{pmatrix}.$$

Thus, $\mathcal{E}_2\mathcal{A}(L)$ is generated by these same terms, and taking the greatest common divisor,

$$\Delta_J^{S^1 \times B^3}(s, t) = s^{-1}t^{-2}.$$

Since $\Delta_J^{S^1 \times B^3}(s, t) \neq \Delta_J(t)$, Lemma 3.1 implies that J is not local in $S^1 \times B^3$.

3.2. Non-Local 2-Knots in Other 4-Manifolds. Armed with the existence of non-local 2-knots in $S^1 \times B^3$, we now return to our original setting: let X be a 4-manifold with $b_1(X) > 0$, and consider a two-component link $L = U \cup J$ in S^4 where U is an unknotted sphere and J is a surface knot. Let E_L denote the complement

$$S^4 \setminus \nu(L) = S^1 \times B^3 \setminus \nu(J).$$

Our eventual goal is to construct a non-local 2-knot in X , but first, we need to know how to relate the Alexander invariants of J in $S^1 \times B^3$ (equivalently of the link L) to locality in

X . Suppose that $\pi_1(E_L)$ and $\pi_1(X)$ have presentations

$$\pi_1(E_L) = \langle \mu, x_1, \dots, x_n \mid r_1, \dots, r_k \rangle \quad \text{and} \quad \langle \alpha, y_1, \dots, y_m \mid s_1, \dots, s_\ell \rangle = \pi_1(X),$$

where μ is represented by the core of $S^1 \times B^3$ (equivalently the meridian of U) and α is a distinguished generator with $\mathfrak{a}(\alpha)$ having infinite order where, again,

$$\mathfrak{a} : \pi_1(X) \rightarrow H_1(X)$$

is the Hurewicz map. Choose a smooth representative γ of $\mathfrak{a}(\alpha)$ in X . If we take a tubular neighborhood of γ , then we can choose an embedding $j : S^1 \times B^3 \hookrightarrow X$ where we identify $\nu(\gamma)$ with $S^1 \times B^3$. After restricting j , we also have an embedding of J into X — this will be our general strategy of transporting non-local 2-knots in $S^1 \times B^3$ to X . Let X_J denote the complement $X \setminus \nu(j(J))$. Note that $j(\mu) = \gamma$ by construction. By Seifert-Van Kampen, $\pi_1(X_J)$ is presented by

$$\langle \alpha, \mu, x_1, \dots, x_n, y_1, \dots, y_m \mid \alpha\mu^{-1}, r_1, \dots, r_k, s_1, \dots, s_\ell \rangle,$$

which is isomorphic to

$$\langle \mu, x_1, \dots, x_n, y_1, \dots, y_m \mid r_1, \dots, r_k, s_1, \dots, s_\ell \rangle,$$

where now in each relation r_i and s_j , we identify α with μ . In order to use Alexander invariants to obstruct locality in X , we would like to take some kind of universal abelian cover of the exterior of $j(J)$ with respect to μ . Intuitively, we do so by ignoring the ambient topology of X . Consider the quotient

$$\mathcal{Q} = \pi_1(X_J) / \langle y_1, \dots, y_m \rangle.$$

Define the map φ to be the composition

$$\begin{array}{ccccc} \pi_1(X_J) & \xrightarrow{q} & \mathcal{Q} & \xrightarrow{\sim} & \pi_1(E_L) \\ & & \searrow \varphi & & \end{array}$$

where q is the quotient map and the final embedding is given by the obvious identification of the generators of the quotient. This is where we use that $\mathfrak{a}(\mu)$ has infinite order, or, more generally, that $b_1(X) > 0$. Each relation s_j in $\pi_1(X)$ is an expression of y_i factors and μ factors. Thus, in the quotient, these relations can only induce some torsion on μ in $\mathcal{Q}$. But since $\mathfrak{a}(\mu)$ has infinite order, we find that $\mathcal{Q}$ is isomorphic to $\pi_1(E_L)$ via φ . Now, to construct our abelian cover, consider the abelianization map

$$\mathfrak{b} : \mathcal{Q} \xrightarrow{\sim} \pi_1(E_L) \rightarrow H_1(E_L) \cong \mathbb{Z}^2.$$

The subgroup $\ker(\mathfrak{b}) = [\mathcal{Q}, \mathcal{Q}]$ of $\pi_1(X_J)$ induces an infinite abelian cover X_L^∞ , and as before, $H_1(X_L^\infty)$ is a Λ_2 -module. Additionally, if $p : X_L^\infty \rightarrow X_J$ is the covering map and $b \in X_J$ is a base point, $H_1(X_J^\infty, p^{-1}(b))$ also has the structure of a Λ_2 -module.

Definition 3.2. The *Alexander module* $\mathcal{A}(J, X, \gamma)$ of J with respect to the pair (X, γ) is defined to be $H_1(X_L^\infty, p^{-1}(b))$ as a Λ_2 -module. Similarly, we define the *Alexander ideals* $\mathcal{E}_i \mathcal{A}(J, X, \gamma)$ and *Alexander polynomial* $\Delta_2 \mathcal{A}(J, X, \gamma)$ which we will denote $\Delta_J^X(s, t)$.

Note that in our notation $\Delta_J^X(s, t)$, we suppress the choice of γ . This is due to the following proposition.

Proposition 3.3. $\mathcal{A}(J, X, \gamma)$ is isomorphic to $\mathcal{A}(L)$ as a Λ_2 -module for any choice of γ in X , and so

$$\Delta_J^X(s, t) = \Delta_J^{S^1 \times B^3}(s, t) = \Delta_L(s, t).$$

Proof. This follows immediately from the fact that $\mathcal{Q}$ is isomorphic to $\pi_1(E_L)$. The presentation matrix we might compute for $\mathcal{A}(J, X, \gamma)$ using the free calculus method is precisely the one we would find for $\mathcal{A}(L)$. $\square$

Now, we apply the same method as before to obstruct locality.

Lemma 3.2. *If $K \subset B^4 \subset X$ is a local surface knot, then $\Delta_K^X(s, t) = \Delta_K(t)$.*

Proof. Since K is local in X , it is also local in $S^1 \times B^3$. Namely, if $K \subset B \subset X$ where B is a 4-ball, then we can choose a smaller 4-ball $B' \subset B$ which is entirely contained in $S^1 \times B^3$. The result follows from Lemma 3.1 and Proposition 3.3. $\square$

Lemma 3.3. *If K is a local surface knot in X , then $J \natural_\alpha K$ is non-local in X , where α is an arc as in Proposition 2.1.*

Proof. Since K is local, we can isotope it to be entirely contained in the image $j(S^1 \times B^3)$. The result then follows from Proposition 3.2 and Lemma 3.2. $\square$

3.3. Proof of Theorem 1.1. Now we summarize and prove the main results of this paper. Let Σ be a compact connected surface, and consider the 2-knot $J \subset S^1 \times B^3$ as constructed in 3.1.

Theorem 3.1. *There exists a non-local null-homotopic 2-knot in X .*

Proof. Choose γ and $j : S^1 \times B^3 \hookrightarrow X$ as before. Since $\pi_2(S^1 \times B^3) = 1$, J is null-homotopic in $S^1 \times B^3$, hence $j(J)$ is also null-homotopic in X . Moreover, this 2-knot in X is non-local by Lemma 3.2, since

$$\Delta_J^X(s, t) = \Delta_L(s, t) = s^{-1}t^{-2}$$

is not a Laurent polynomial in t and cannot equal $\Delta_J(t)$. $\square$

Proof of Theorem 1.1. Choose some local surface knot K in X which is an embedding of Σ . Using Theorem 3.1, let J be a null-homotopic non-local 2-knot in X , and choose an arc α as in Proposition 2.1 between J and K . Then by Lemma 3.3, the surface knot $K \natural_\alpha J$ is also a non-local surface knot which is an embedding of Σ . Moreover, $K \natural_\alpha J$ is null-homotopic, since both K and J are null-homotopic. $\square$

Corollary 1.1 follows from a similar argument with a slight change in setup.

Proof of Corollary 1.1. Let M be a compact 3-manifold with $b_1(M) > 0$. Then $M \times I$ is a 4-manifold with $b_1(M) > 0$. If K is a knot which is slice in B^4 , then we can find a slice disk D in $M \times I$. Once again, let J be a non-local 2-knot in $M \times I$ and α an arc as in Proposition 2.1. Then $J \natural_\alpha D$ is also a slice disk for K , and by the same argument as above, $J \natural_\alpha D$ is non-local. $\square$

4. FURTHER QUESTIONS

A natural generalization of these results would be to show there exists non-local surface knots in a 4-manifold X with less restrictions on its algebraic topology.

Question 4.1. *if X has $H_1(X) \neq 0$ but $b_1(X) = 0$, does there exist non-local surface knots? Can a similar argument work to the one presented above?*

It may be possible to treat the cases where X has $b_1(X) = 0$ but $H_1(X) \neq 1$ with a similar strategy — for example by choosing a distinguished generator α of $H_1(X)$ and embedding $S^1 \times B^3$ around a neighborhood of a smooth representative of α — but it is unlikely that the method of Alexander invariants would lead to a fruitful outcome. In this case, the quotient $\mathcal{Q}$ as in 3.2 would *not* be isomorphic to $\pi_1(E_L)$, but instead a subgroup of $\pi_1(E_L)$ where μ has torsion. Then the abelianization of $\mathcal{Q}$ lands in $\mathbb{Z} \oplus \mathbb{Z}/n$ where $\mu^n = 1$, yielding the structure of a $\mathbb{Z}[\mathbb{Z} \oplus \mathbb{Z}/n]$ -module. Unfortunately, the ring $\mathbb{Z}[s^{\pm 1}, t^{\pm 1}]/(t^n - 1)$ is not even a UFD, so such invariants are likely not as helpful as one might hope. We do conjecture that this construction gives non-local 2-knots in X , but this is not something Alexander invariants can readily detect.

Question 4.2. *If X has $\pi_1(X) \neq 1$ but $H_1(X) = 1$, does there exist non-local surface knots?*

In this case, invariants coming from abelianization are even less helpful. One would need to obstruct locality at the level of $\pi_1(X)$, or demonstrate a more geometric argument for non-locality. One potential approach to this problem in the context of slice disks in particular comes from a variant on the construction of classical null-homotopic non-local knots. Let M be a 3-manifold which is not S^3 , and let K be a local knot in M which is slice in B^4 . Following the answer to Question 39 in [Cel18], we see that K is concordant to a non-local knot K' , in particular a K' which is slice in B^4 . Suppose for contradiction that the annulus A in $M \times I$ which cobounds K and K' is local. It may be possible to show that the 4-ball around A contradicts the locality of K' , proving Corollary 1.1 for X with $H_1(X) = 1$.

It is also worth noting that many of the arguments presented here seem to generalize to knots of any dimension $n \geq 2$. Once one proves that non-local n -knots exist in $S^1 \times B^{n+1}$, a similar Alexander invariant argument can be run to show the existence of non-local n -knots in any $(n+2)$ -manifold X with $b_1(X) > 0$. To show the former, it may be possible to iterate the spun knot construction to get non-local n -knots in $S^1 \times B^{n+1}$.

Lastly, a question closer in motivation to Corollary 1.1 that would be particularly interesting to explore is:

Question 4.3. *If M is a 3-manifold which is not S^3 and $K \subset B^3 \subset M$ is a knot which is slice in B^4 , does there exist a slice disk D which is topologically but not smoothly isotopic to a local slice disk $D' \subset B^4 \subset M \times I$?*

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